

Representation Adequacy and Transportable Relational Profiles in Omics Clustering

TRPP method family - detailed project description SONATA BIS 16 | 48 months | primary panel ST6 Scientific draft: complete in scope and logic. Verified personnel, institutional, budget, and publication-record data must be entered in OSF by the applicant and host institution.

1. Scientific objective

1.1 Problem

Clustering is widely used to seek molecularly distinct patient groups, but its first consequential choice is often hidden: the representation in which similarity is calculated. A patient may be described by normalised measurement values, within-sample ranks, binary or ternary ordering relations, or a mixture of these views. Each representation encodes a different invariance. Value geometry retains magnitude but can be sensitive to calibration and scale. Rank-based geometry discards magnitude and is invariant to sample-wise strictly monotone transformations, yet it can be unstable around ties. Pair relations such as gene A > gene B are directly readable and potentially transferable, but gene-specific shifts, missing features, and annotation changes can reverse them. A hybrid can exploit complementary information but can also fit noise if its weight is selected retrospectively.

Most workflows select one representation by convention and then optimise the clustering algorithm. This conflates geometry, grouping, and interpretation. Moreover, any clustering engine can produce a partition even when the data do not contain a reproducible group structure. A stable discovery partition may track a technical nuisance, and a partition that is stable within one cohort may fail when assigned to new patients. These limitations are especially important in omics studies, where dimensionality is high, cohorts are modest, and platforms and populations differ.

The project asks one central question: under which source-observable conditions should magnitude information be retained, replaced by within-sample relations, combined with them, or judged insufficient for stable and transportable omics clustering? A source-only Representation Audit first determines whether VALUE, RELATIONAL, HYBRID, or NO_STABLE_STRUCTURE is supported. For eligible relational cases, the project will develop Transportable Relational Patient Profiles (TRPP) as sparse regions: short sets of within-patient rules that define a group and can assign a new patient without reclustering the target cohort. The entire artifact is frozen before target evaluation, and insufficient evidence yields UNASSIGNED.

1.2 Main hypothesis

Measurable signal and shift properties determine whether transportable grouping should retain values, use within-sample relations, combine both views, or abstain; within the relational domain, biologically useful patient groups can be represented by sparse frozen rule sets.

This is not a hypothesis that relations are universally superior. It predicts distinct competence regimes and permits value geometry or abstention to be the correct outcome.

1.3 Specific hypotheses and objectives

H1 - representation adequacy. Source-only stability, prediction strength, non-degeneracy, matched-null evidence, and explicit invariance properties can distinguish competence domains of VALUE, RELATIONAL, and HYBRID and detect absence of stable structure. The corresponding objective is to build and validate a representation-adequacy map, not merely a winner-take-all selector.

H2 - direct relational regions. Direct joint learning of assignments and sparse relational profiles can provide equal or better external transfer than relational PAM followed by post-hoc profile extraction while improving profile stability, concision, or both. The objective is a deterministic algorithm with core/optional relations, redundancy control, and a fully replayable trace.

H3 - frozen transport. A source-learned profile can retain non-trivial coverage and external associations in independent cohorts without target refitting, joint normalisation, or target-guided feature replacement. The objective is to test this claim in two disease modules with at least three cohorts each.

H4 - applicability boundaries. Signal type, noise, missing features, platform mapping, and biological composition generate predictable conditions in which relations help, remain neutral, or fail. The objective is a quantitative map connecting controlled perturbations with observed multi-cohort failures.

The four objectives are:

1. establish a leakage-controlled Representation Audit with an explicit NO_STABLE_STRUCTURE outcome;
2. develop and gate direct sparse relational regions against a strong post-hoc baseline;
3. evaluate completely frozen profiles in independent lung and colorectal cancer cohorts;
4. release an empirical domain-of-applicability map and deterministic reference implementation.

1.4 What the project will not claim

The project is basic methodological research. It will not claim that unsupervised groups are diagnoses, that stability proves biological validity, or that a profile is clinically useful without prospective validation. Target labels cannot train, select, or rescue a method. Fuzzy clustering, evolutionary optimisation, deep learning, federated learning, CUDA, and a web portal are outside scope. Anchor sets are outside the core scope; a single-anchor restriction is only a conditional efficiency experiment after the direct- region gate.

2. Significance and innovation

2.1 Scientific need

Cluster discovery is unusually vulnerable to researcher degrees of freedom. Unlike supervised prediction, it offers no training outcome against which a representation can be judged directly. Stability methods test persistence under perturbation, prediction strength asks whether a partition learned on one split can be reproduced on another, and external indices compare a frozen partition with later labels. These address different questions. A coherent workflow must keep them separate, expose the null case, and demonstrate which data were available at every decision.

2.2 State of the art and conceptual gap

PAM is a natural common engine for comparing representations because it accepts an arbitrary dissimilarity matrix and represents each group by an observed medoid rather than a synthetic mean. However, PAM does not determine which dissimilarity is scientifically appropriate and will return a partition under null structure. Consensus and resampling stability methods quantify reproducibility, but high stability can be driven by a technical or biological nuisance unrelated to the phenotype later considered important. Prediction strength formalises train-to-test reproducibility but does not create a sparse, transportable group definition.

Within-sample ordering approaches offer attractive invariance and interpretability. Top-scoring-pair and related classifiers demonstrated that a small number of relative expression comparisons can support transparent supervised decisions. Their objective and validation differ fundamentally from unsupervised patient stratification: class labels select the pairs, while a cluster profile must emerge without such labels. A post-hoc profile can explain a given cluster, but explanation after clustering does not show that the rules were sufficient to define or transfer the group.

Multi-view and multiple-kernel methods already combine representations, and interpretable clustering already learns tree- or rule-based descriptions. Consequently, neither view fusion nor rules alone constitute the proposed novelty. A fitted hybrid can also benefit from target or label information if weights are selected after evaluation. Batch correction addresses a related but different problem. Methods such as ComBat can enable joint analysis when batches are explicitly modelled; they are not appropriate for the project's primary frozen-transfer claim if fitting them uses the target cohort distribution. The project therefore compares cohort-wise preprocessing and source-frozen mapping in the primary analysis, with pooled batch correction only as a clearly separated secondary analysis that cannot support a frozen- transfer claim.

2.3 Innovation

The innovation lies in the architecture and optimisation target:

- Adequacy before grouping: multiple geometries are audited under one deterministic clustering engine, with component diagnostics and a null- calibrated abstention option.
- Direct rather than retrospective regions: sparse relations participate in defining assignments, instead of being selected only to describe a completed partition.
- Two forms of abstention: an analysis may report NO_STABLE_STRUCTURE, and an individual target sample may be UNASSIGNED.
- Frozen assignment: each target sample is evaluated using an immutable source artifact, and its assignment cannot depend on other target samples.
- Applicability as a scientific result: failures caused by gene-specific shifts, missing mapping, nuisance, or mixture change are measured and synthesised rather than removed from the report.
- Executable evidence boundary: label access, artifact hashes, environment, decisions, and GO/STOP transitions are machine-checkable.

This combination converts a vague promise of robustness into falsifiable claims about what is learned, where it applies, and when it must abstain.

3. Preliminary results and starting point

3.1 Implemented pilot infrastructure

The completed pilot is a Python-first, deterministic, CPU-only reference pipeline. It includes a `DatasetBundle` separating measurements from labels; a source-only preprocessing artifact; average ranks with explicit ties; ternary pair encoding; value Euclidean/correlation, rank-footrule, pair-Hamming, and hybrid distances; and one deterministic PAM implementation with golden tests. The audit layer implements perturbation stability, prediction strength, non-degeneracy checks, matched NULL calibration, finite-grid representation selection, frozen target assignment, and immutable prelabel artifacts. Fit and audit modules cannot import the label loader. Correctness, determinism, and leakage barriers were tested before larger stages proceeded.

3.2 Controlled benchmark

The primary simulation grid contained 630 independently evaluated source-target pairs across VALUE, RELATIONAL, HYBRID, and NULL regimes, multiple signal strengths, and domain shifts. The audit identified the generating representation family in 0.933 of signal replicates. Median target ARI regret relative to the retrospectively best candidate was 0.000. The false-structure rate under NULL was 0.067, and HYBRID was selected in 0.000 of pure VALUE or RELATIONAL regimes. Differences in source-only audit quality were associated with corresponding differences in target behaviour (Spearman $\rho = 0.854$). Every preregistered Gate B criterion passed without changing its threshold. Gate B: GO.

3.3 Real-data audits

Eleven heterogeneous expression datasets were then audited within dataset. The complete decision and assignments for every dataset were written to immutable, hashed prelabel artifacts before labels were loaded. RELATIONAL was selected in eight datasets, VALUE in two, and HYBRID in one. The selected method lay within 0.05 ARI of the retrospective within-dataset oracle in 9/11 datasets, with median regret 0.011. Yet median ARI against available labels was only 0.065. This is descriptive evidence, not external validation. It demonstrates that finding a candidate geometry close to the best available geometry and finding a partition that matches a particular phenotype are empirically distinct tasks.

3.4 Frozen external transfer and its consequence

Bidirectional transfer was evaluated between independent lung-expression cohorts GSE10072 and GSE19804. Each direction selected 200 source features from a predeclared common universe. Source preprocessing, candidate relations, distance scales, representation, medoids, and rejection thresholds were frozen before target labels were evaluated; target samples were assigned without refitting or joint normalisation.

GSE19804 to GSE10072 selected pair-Hamming and achieved target ARI 0.926, matching the retrospective oracle with regret 0.000. GSE10072 to GSE19804 selected value-correlation and achieved ARI 0.559 versus oracle 0.664, a regret of 0.105284. The frozen reverse-direction allowance was 0.10. Gate C: STOP. No threshold, feature, relation margin, weight, or selector was changed. PILOT-016 to PILOT-018 were not run; direct regions: NOT TESTED; anchors: NOT TESTED. The labels were used only after all source-only decisions and assignments had been frozen.

The failed gate prevents an inflated preliminary claim and strengthens the project logic. It shows that controlled adequacy is promising but insufficient, and identifies transfer asymmetry as an object requiring multiple cohorts, explicit coverage, and direct profile learning.

4. Research plan

WP1 - Representation adequacy, invariance and clustering abstention (months 1-12)

WP1 will formalise an operational invariance and information-loss taxonomy for the finite VALUE, rank, ternary-relation, and HYBRID family, and will extend controlled regimes. It will state which transformations preserve each view and which remove or reverse information, then test those predictions. Generators will separate value-magnitude signal, stable relative-order signal, complementary hybrid signal, and no structure. Orthogonal perturbations will include global and feature-specific shifts, monotone sample transformations, variance changes, dropout and missing features, unbalanced groups, nuisance factors, mapping errors, and source-target mixture changes. Unknown numbers of groups will be included after the fixed-K correctness layer is locked.

Every primary matched comparison uses the same deterministic PAM. Candidate cluster numbers and hybrid weights are finite and preregistered. For each candidate, the source-only audit records perturbation stability, prediction strength, minimum group fraction, empty/duplicate-medoid conditions, assignment margin, and excess over a matched permuted or simulated NULL. A candidate is eligible only if all hard constraints pass; an aggregate quality score ranks eligible candidates. If none qualifies, the output is NO_STABLE_STRUCTURE.

Milestones: M1, expanded deterministic generators and unit tests (month 4); M2, frozen audit definition and null calibration (month 8); M3, preregistered controlled benchmark and applicability summary (month 12).

Gate G1: predefined minimum family discrimination, maximum NULL false-structure rate, maximum unjustified HYBRID rate, low target regret, and positive source/target association must be met on held-out controlled regimes. If the winner-take-all choice is not supported but diagnostics remain informative, the scientific output becomes an adequacy vector plus abstention. Target evidence cannot recalibrate the primary gate.

WP2 - Direct sparse relational regions (months 7-24)

WP2 starts with an explicit RR_POSTHOC baseline: relational PAM creates a partition, after which relations that are frequent within and contrasting between groups are selected. This baseline isolates the benefit of direct learning from the benefit of using relations at all.

RR_DIRECT will jointly update assignments and sparse profiles. A profile contains relation identifiers, expected ternary states, source frequencies, weights, core/optional status, missingness allowances, and a calibrated assignment threshold. The objective balances within-profile agreement, between-profile separation, coverage, number of relations, number of distinct genes, and redundancy. Inverse duplicates are forbidden. Limits on reuse of one gene prevent superficially short profiles dominated by one measurement. Candidate profile lengths 10, 25, and 50 provide interpretable operating points.

Initialisation, candidate ordering, tie-breaking, empty-group repair, and stopping are deterministic. Every iteration writes an audit trace. Samples may remain temporarily or finally unassigned if no profile has adequate observable support and margin.

WP2 validation and gate

Correctness tests will cover ternary ties, missing relations, inverse relation removal, objective monotonicity where required, deterministic ties, empty groups, and equality between repeated runs. Synthetic recovery experiments will measure known region recovery in addition to partition agreement. Bootstrap analysis will match relations by canonical identifiers and report Jaccard, frequency, state consistency, and gene-level stability.

RR_DIRECT is compared with RR_POSTHOC under identical source features, relation budgets, cluster numbers, and transfer tasks. Gate G2 passes if direct learning improves external ARI/NMI by at least 0.05, improves bootstrap profile Jaccard by at least 0.10, or remains within 0.03 of the baseline while producing a materially shorter profile. The exact definition of materially shorter will be frozen from source-side simulations before real target labels are opened. If no criterion passes, direct development stops; the negative result and post-hoc baseline remain deliverables.

Milestones: M4, tested post-hoc baseline (month 12); M5, traceable direct prototype (month 18); M6, controlled and source-resampling evaluation (month 21); M7, G2 decision (month 24).

WP3 - Frozen multi-cohort TRPP transfer (months 18-38)

WP3 uses two modules chosen to expose complementary transfer problems.

Lung adenocarcinoma. GSE10072 is the default discovery cohort: 107 final GPL96 profiles, including 58 tumours and 49 non-tumour tissues. GSE19804 provides 120 GPL570 profiles from 60 paired cancers and adjacent normal tissues in non-smoking women from Taiwan. GSE32863 provides 116 GPL6884 Illumina profiles from 58 matched lung adenocarcinomas and adjacent non-tumour tissues. GSE18842 is a declared sensitivity cohort because it includes broader NSCLC histology. The primary question is not whether tumour/normal can be classified, but whether the source-supported geometry and concise profile remain applicable under etiological, population, and platform shifts.

Colorectal cancer. Tumour samples from GSE39582 form the default discovery cohort. GEO records 585 GPL570 profiles, including 443 discovery tumours, 123 internal validation tumours, and 19 non-tumour mucosa samples. Independent targets are GSE14333, with 290 primary colorectal tumours, and GSE33113, with 90 stage-II tumours plus six matched normal samples. The primary tumour analysis does not use normal samples to create an easy contrast. Published CMS labels, stage, molecular markers, recurrence, and survival are loaded only for final evaluation. Because CMS is expression-derived, it is an external reference, not a label-independent biological truth.

GSE39582's internal split is not counted as an independent study. Every cohort must pass a sample-level metadata, duplicate, assay, scale, and annotation audit before the multi-cohort protocol is preregistered.

WP3 frozen-transfer protocol

For each module, a discovery analysis fits all preprocessing and modelling objects on the source cohort. It records sample exclusions, source feature selection, identifier mapping, probe-to-gene aggregation, candidate relations, distance normalisation, audit decision, cluster number, medoids or regions, profile weights, coverage requirements, assignment margins, and rejection thresholds. The artifact contains ordered identifiers, code revision, environment, seeds, and cryptographic hashes.

Target cohorts are processed separately. No joint source-target normalisation, target distribution fitting, target-label feature selection, or target-guided replacement of missing relations is allowed in the primary analysis. Platform mapping follows the frozen annotation rule. If coverage is below the declared threshold, the sample is UNASSIGNED; if the task cannot support the profile, the task is reported ineligible. Assigning one sample alone and within a target batch must give the same result.

Primary baselines are VALUE PAM, rank-footrule PAM, pair-Hamming PAM, finite- grid HYBRID PAM, and RR_POSTHOC, all using the same source split and preprocessing boundary. RR_DIRECT enters only after G2. A conventional cluster-then-classify pipeline and pooled harmonisation may be secondary comparators, clearly labelled as answering different questions. They cannot be used to validate frozen transfer.

All declared source-to-target directions are reported. Reverse directions are secondary and preregistered only where sample composition makes them scientifically interpretable. A poor direction is not averaged away. Dataset exclusions made after label inspection are prohibited.

Milestones: M8, final metadata/mapping audit and preregistration (month 22); M9, lung prelabel artifacts (month 28); M10, colorectal prelabel artifacts (month 32); M11, locked evaluation and cross-module synthesis (month 38).

WP4 - Applicability map, conditional efficiency, and release (months 31-48)

WP4 will link controlled perturbations with observed cohort characteristics. The map will describe risk as a function of source stability, NULL margin, feature/relation coverage, relation flip rate, assignment margin, group balance, platform mapping, and metadata shift. It will distinguish four conclusions: supported relational transfer, supported value transfer, complementary hybrid transfer, and insufficient evidence.

Only if G2 passes, a single-anchor restriction may test whether relations anchor > feature reduce computation while retaining performance. It must achieve at least a fivefold cost reduction, ARI loss no greater than 0.03, and no material coverage or stability loss. Anchor sets and evolutionary searches remain excluded.

Milestones: M12, applicability model (month 42); M13, conditional efficiency decision (month 44); M14, reference software, manifests, benchmark, and final synthesis (month 48).

5. Methods

5.1 Data objects and leakage firewall

Each cohort is imported into a `DatasetBundle` containing the measurement matrix, feature identifiers, sample identifiers, and non-outcome technical metadata. Evaluation labels are stored in a separate file and exposed through a dedicated evaluation package. Fit and audit packages are tested to ensure they cannot import the label loader. Depending on the preregistered question, clinically descriptive variables that could reveal outcomes are also kept on the evaluation side.

Source preprocessing produces an immutable artifact. For arrays, the project will prefer raw-data cohort-wise preprocessing when raw files and annotation permit it; otherwise it will document the deposited processed scale and perform predeclared quality checks. Probe filtering and probe-to-gene rules are fitted or fixed on source. Robust scaling for VALUE uses source location and scale. Feature selection uses source-only dispersion, normally median absolute deviation, with stable identifier tie-breaking. Target values never update these parameters.

5.2 Representations and distances

For a sample vector with possible ties, ranks use the average-rank convention. A pair relation for features j and k is $+1$ if x_j exceeds x_k , -1 if it is lower, and 0 for an observed tie. Missingness is a separate mask and cannot be silently encoded as zero. Candidate relations are screened from source by predeclared prevalence, contrast, and redundancy rules without labels.

VALUE uses source-scaled Euclidean distance and correlation distance where defined. RELATIONAL uses normalised Spearman-footrule distance between average ranks and Hamming-type disagreement among observed ternary relations. HYBRID is a convex combination after each component is scaled by a source-only distance functional. Candidate weights form a finite frozen grid. Each distance is tested for symmetry, diagonal zero, bounds where applicable, missingness handling, and deterministic serialisation.

5.3 Deterministic PAM

All matched representations use the same PAM implementation. BUILD and SWAP steps have canonical sample-identifier tie-breaking, explicit duplicate and non-finite distance rejection, deterministic empty-group protection, and a recorded objective trace. Golden tests enumerate all medoid combinations on small matrices and confirm the global optimum. Repeated-process and permuted- input tests verify stable canonical output. This design ensures that a representation comparison is not an accidental comparison of clustering engines.

5.4 Source-only audit

Perturbations are generated from source without labels and include patient subsampling, feature subsampling, bounded measurement noise, and relation dropout. Stability is measured after matching partitions with ARI and pairwise co-membership summaries. Prediction strength estimates reproducibility from source train to source validation splits. Hard eligibility checks include minimum group fraction, absence of duplicate medoids, finite distances, and excess over a matched NULL threshold. All component values are retained. The quality score and selection rule are frozen before external evaluation.

5.5 Relational-region model

Let a profile r contain a subset of relations S_r , expected states s_r , weights w_r , core indicators c_r , and source-derived missingness allowances. For sample x , its observable agreement with r is the weighted agreement over relations present in x , penalised when core coverage is insufficient. Assignment uses the best score only when total coverage, core coverage, absolute score, and margin over the second-best profile exceed frozen thresholds. Otherwise x is UNASSIGNED.

RR_POSTHOC selects S_r after PAM from within-group prevalence and between-group contrast. RR_DIRECT optimises assignments and S_r jointly through deterministic coordinate updates. Candidate additions and deletions are ordered canonically; ties are resolved by ordered relation identifiers. The objective includes compactness, contrast, coverage, profile length, inverse redundancy, and gene reuse. The algorithm stops when no admissible update improves the objective beyond a fixed tolerance or a fixed iteration limit is reached. The trace makes every change replayable.

5.6 Evaluation outcomes

For controlled data, primary outcomes are exact representation-family identification, target ARI and NMI, target ARI regret versus the retrospective oracle, NULL false-structure rate, HYBRID selection in pure regimes, and Spearman association between source audit differences and target-performance differences. Known generative regions add relation precision/recall and state recovery.

For real cohorts, the project reports:

- eligibility, assignment coverage, UNASSIGNED fraction, and minimum group size;
- profile length, number of genes, observable relation fraction, core coverage, relation flip rate, and assignment margins;
- bootstrap relation Jaccard and state stability;
- frozen ARI/NMI against declared external references where appropriate;
- clinicopathological, molecular, pathway, recurrence, and survival associations as evaluation-only outcomes;
- performance and uncertainty for every preregistered cohort and direction.

ARI/NMI quantify agreement, not biological truth. Survival analyses will use predeclared endpoints and proportional-hazards diagnostics; effect sizes and confidence intervals will accompany p-values. Molecular and pathway tests will use multiplicity control. Paired specimens, repeat measures, and technical replicates are resampled at patient level. Cohort-level estimates will be shown individually before any hierarchical or meta-analytic summary.

5.7 Robustness and negative controls

Negative controls include permuted samples/features, matched NULL generators, label permutation confined to evaluation, synthetic nuisance structure, and profiles applied after deliberate feature removal. Sensitivity analyses vary only factors declared before target-label access: source feature budget, relation budget, profile size, cluster number range, and mapping rule. Primary results remain tied to the preregistered configuration.

6. Data modules and validity threats

6.1 Lung module

GSE10072, GSE19804, and GSE32863 contain lung tumour and adjacent/non-tumour expression profiles but differ in platform, population, smoking context, and sample pairing. This makes them informative for transport and dangerous for causal interpretation. Cohort identity is a composite shift; success cannot be attributed to a single technical or biological factor. The core analysis uses histologically aligned lung adenocarcinoma where metadata permit. GSE18842 is a broader-NSCLC sensitivity analysis because mixing squamous and adenocarcinoma could create an easy histology partition.

The tumour/non-tumour signal is intentionally treated as a stress-test scaffold rather than the ultimate biological discovery. After the representation and transfer machinery is verified, tumour-only exploratory profiles may be evaluated under a separately preregistered question if sample size and metadata support it. They cannot replace a failed primary task.

6.2 Colorectal module

GSE39582 offers a large discovery series with molecular and clinical annotation. GSE14333 and GSE33113 offer independent patient series but differ in stage composition and original processing. The analysis therefore focuses on coverage and retention of frozen tumour profiles rather than forcing identical source group proportions. Normal tissue is excluded from the primary tumour- subtype task. The internal GSE39582 validation subset is valuable for source development but is not counted as an independent cohort.

CMS provides a biologically interpretable consensus reference, yet it is built from gene expression. Agreement with CMS may therefore reflect shared signal rather than independent truth. Stage, MSI/MMR, mutations, location, treatment, recurrence, and survival provide complementary evaluation where complete. None can select the representation or profile.

6.3 Threats to validity and mitigation

Leakage: joint scaling, feature selection, batch correction, or threshold selection would invalidate frozen transfer. Mitigation is a physical label firewall, immutable artifacts, import tests, and target-mutation invariance tests.

Annotation instability: probe-to-gene changes may alter orderings. Mitigation is a versioned mapping, source-fixed aggregation, coverage reporting, and a sensitivity analysis across independently declared annotation versions.

Confounding: cohort, platform, population, histology, and treatment may be inseparable. Mitigation is per-cohort reporting, declared sensitivity cohorts, controlled perturbations, and restrained causal language.

Selective reporting: a method can appear transferable if only favourable directions are retained. Mitigation is preregistration, complete task manifests, and publication of ineligible and failed tasks.

Circular biological evaluation: expression-derived subtype labels share information with expression clusters. Mitigation is explicit classification of CMS as a reference, separate clinical endpoints, and no label-guided learning.

7. Project governance, reproducibility, and data management

7.1 Decision governance

Every WP has a frozen analysis specification and an explicit transition. Threshold changes are allowed only before the relevant target evaluation and must create a new version; they cannot rewrite a completed primary result. Gate failure changes downstream scope exactly as declared. The decision log records who approved a change, which evidence was available, and which artifacts became obsolete. Negative results are retained.

7.2 Software and computational reproducibility

The reference implementation is Python-first and CPU-only. Dependencies are pinned, environments are recorded, and pseudorandom components use explicit seeds and deterministic ordering. Canonical JSON serialisation supports byte-level comparison of model artifacts. Continuous tests cover numerical correctness, determinism across processes, label import boundaries, target mutation invariance, individual/batch assignment equivalence, and schema validation. Larger benchmark stages start only after their acceptance suite passes.

Public releases will contain source code, synthetic generators, configuration schemas, dataset accession manifests, annotation hashes, tests, and aggregated results. Raw public repositories will be referenced rather than redistributed when their terms or size make redistribution inappropriate. Derived artifacts will avoid direct identifiers. Each release will include environment and provenance metadata.

7.3 Data management and ethics

The default datasets are public, de-identified omics cohorts. Before use, the team will review repository terms, consent restrictions, controlled-access requirements, and the host institution's ethics guidance. Only data necessary for the declared research questions will be processed. Local working copies will be access-controlled and integrity-checked. Public outputs will contain code, non-identifying manifests, model profiles, and aggregate results unless the original terms impose stricter limits.

The method is not a medical device and will not produce treatment advice. Profiles will be presented as research hypotheses. Any later prospective or clinical study would require separate ethical, regulatory, and clinical validation. The final OSF data-management and ethics forms will be completed with the host institution using the verified storage, retention, backup, and responsibility arrangements.

7.4 Open science

The project will preregister primary simulations, cohort roles, gates, and evaluation order. Manuscripts will distinguish confirmatory from exploratory analyses. Code and reproducible artifacts will be released at stable revisions, with licences compatible with source datasets and dependencies. Results will include failures and applicability limits, reducing incentives to report only successful transfers.

8. Team and feasibility

The PI will lead scientific design, relational-learning methodology, gate decisions, supervision, and synthesis. A doctoral researcher employed for at least 36 months will develop RR_DIRECT, tests, and stability theory as a coherent dissertation project. A competitively recruited postdoctoral researcher or specialist analyst will independently curate cohorts, harmonise metadata, verify frozen artifacts, and lead replication. Confirmed external consultants may advise on biological interpretation but will not own the core method or replace independent validation.

8.1 Why a new team is necessary

The work combines three substantial and deliberately separated responsibilities: algorithm development, multi-cohort bioinformatics, and independent reproducibility review. A single investigator performing all three would create a bottleneck and increase confirmation bias. The doctoral project gives depth to direct-region methodology; the postdoctoral/specialist role ensures that cohort curation and target evaluation are not informal extensions of method development. The PI integrates the work and is accountable for all frozen decisions.

The named-team section will be completed only after eligibility, employment, prior joint-project restrictions, and institutional rules are checked. No person's achievements or commitment are inferred in this scientific draft.

8.2 Feasibility

The pilot provides operational infrastructure rather than a mock-up. It already implements the data boundary, source artifacts, representations, distances, deterministic PAM, controlled grids, real-data adapters, frozen assignment, evidence validation, automated reports, and tests of correctness, determinism, and leakage. The complete 630-pair benchmark was executed. The formal STOP at Gate C shows that stopping logic works in practice. The project begins from tested Level-1 to Level-3 components while keeping direct regions as genuine research.

Public cohorts make the 48-month plan realistic without depending on new sample recruitment. The main resource requirement is staff time and deterministic CPU compute. Relation screening and explicit candidate budgets bound the quadratic feature-pair space. Intermediate artifacts permit restart and audit. The two disease modules offer replication without turning the project into a broad disease survey.

9. Risk register and fallback results

1. No representation reliably clears the NULL boundary. Report the conditions and retain abstention as the result; do not lower thresholds using target labels.
2. Source diagnostics predict only a ranking, not an exact winner. Publish an adequacy vector and uncertainty rather than a universal selector.
3. Direct regions do not improve on post-hoc profiles. Stop RR_DIRECT at G2; release the negative comparison and use RR_POSTHOC downstream.
4. Cross-platform coverage is inadequate. Report ineligibility or UNASSIGNED; analyse mapping sensitivity without replacing relations based on target outcomes.
5. Stable profiles lack clinical association. Separate structural from phenotype validity; the negative association is scientifically meaningful.
6. Profiles track technical or cellular-composition nuisance. Use metadata diagnostics, controlled nuisance regimes, pathway/cell-content assessment, and cross-cohort behaviour; avoid causal claims.
7. Computation is excessive. Reduce the source candidate budget by a preregistered unsupervised rule. Test a single-anchor restriction only after G2; do not introduce evolutionary or GPU methods.
8. One disease module becomes unusable because metadata or terms change. A reserve cohort may replace it only after a documented prelabel audit and before any target-label comparison; the original exclusion remains public.

10. Schedule, deliverables, and success criteria

10.1 Timeline

- Months 1-6: recruit team; lock data governance; extend generators and correctness suite; perform accession-to-sample metadata audit.
- Months 7-12: freeze and run WP1 benchmark; calibrate abstention; decide G1; implement RR_POSTHOC.
- Months 13-18: implement direct optimisation, trace, missingness, and core/optional profiles; run synthetic recovery and determinism tests.
- Months 19-24: compare direct and post-hoc profiles; freeze and decide G2; preregister accepted cohort mappings and WP3 tasks.
- Months 25-32: generate lung and colorectal prelabel artifacts; perform independent integrity review before labels are loaded.
- Months 33-38: execute locked evaluation, sensitivity analyses, and complete per-cohort reporting.
- Months 39-44: estimate applicability map; run the conditional single- anchor experiment only if authorised by G2.
- Months 45-48: integrate theory and empirical results; archive the reproducible benchmark and reference implementation.

10.2 Deliverables

- D1: versioned Representation Audit specification, generators, and NULL calibration;
- D2: controlled competence map with G1 record;
- D3: tested RR_POSTHOC baseline;
- D4: RR_DIRECT implementation and G2 comparison, including a negative result if applicable;
- D5: lung module prelabel and evaluation artifacts;
- D6: colorectal module prelabel and evaluation artifacts;
- D7: cross-module applicability map and conditional efficiency decision;
- D8: open Python reference package, documentation, tests, manifests, and final synthesis.

10.3 Criteria for success

Project success is not defined as relational victory on every dataset. It requires: correct and deterministic implementations; demonstrated absence of target/label leakage; calibrated false-structure control; honest operation of G1/G2; at least two fully audited multi-cohort modules unless a documented data eligibility failure intervenes; complete reporting of coverage and negative directions; and a usable map distinguishing supported transfer from abstention.

The strongest positive result would be a concise direct profile that retains assignments and external associations across independent platforms. A valid negative result would show that post-hoc profiles, value geometry, or abstention is preferable under identifiable conditions. Both advance the central goal: making omics clustering more interpretable, falsifiable, and transport-aware.

11. Expected scientific impact

The project will shift emphasis from producing a partition to justifying a frozen patient representation. It will provide researchers with evidence about when a within-sample relational description is credible, when magnitude information must be retained, and when data do not support grouping. Direct regions will connect unsupervised discovery with an interpretable rule-based assignment mechanism, while the applicability map will prevent invariance claims from being generalised beyond observed conditions.

The project also contributes a reproducibility pattern applicable beyond omics: physical label separation, prelabel artifacts, common engines for representation comparisons, explicit null outcomes, and irreversible evaluation gates. The result will be an open research method and evidence base, not a portal or clinical product.

12. References

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